

Chapter 3: The Z-transform

Chapter 4: Frequency Analysis of Signals & Systems

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Prob 1:

Determine the z-transform and ROC:

- a. $x[n] = u[n - 10]$
- b. $x[n] = (-0.8)^n u[n]$
- c. $x[n] = n \cdot 2^n u[n]$
- d. $x[n] = (-0.8)^n (u[n] - u[n - 10])$
- e. $x[n] = 2((0.8)^n u[n] + (-0.8)^n u[n])$
- f. $x[n] = 4(u[n] + (-1)^n u[n])$
- g. $x[n] = \cos(\pi n) u[n]$
- h. $x[n] = \frac{1}{2}[(0.8j)^n u[n] + (-0.8j)^n u[n]]$
- i. $x[n] = \begin{cases} \frac{1}{3} & n \geq 1 \\ 2 & n < 1 \end{cases}$

Prob 2:

Determine the z-transform and ROC:

- a. $x[n] = (0.8)^n u[n] + (1.25)^n u[n]$
- b. $x[n] = (0.8)^n u[n] - (1.25)^n u[-n - 1]$
- c. $x[n] = (0.8)^n u[-n - 1] - (1.25)^n u[-n - 1]$

Prob 3:

Determine the inverse z-transform:

- a. $X(z) = \frac{1}{4} \cdot \frac{z^2}{(z-1)(z+3)}$
- b. $X(z) = 5 + 3z^{-1} + 2z^{-2}$
- c. $X(z) = \frac{5}{1-0.64z^{-2}}$

- d. $X(z) = \frac{10+z^{-1}}{1+0.25z^{-2}}$
- e. $X(z) = \frac{6-3z^{-1}-2z^{-2}}{1-0.25z^{-2}}$
- f. $X(z) = -4 + \frac{1}{1+4z^{-1}}$
- g. $X(z) = \frac{3(1+0.3z^{-1})}{1-0.81z^{-2}}$
- h. $X(z) = \frac{z^{-1}+2z^{-2}+3z^{-3}+4z^{-4}}{1-z^{-4}}$
- i. $H(z) = \frac{(2-3z^{-1})(1+z^{-1})}{1-0.25z^{-4}}$

Prob 4:

Determine system function $H(z)$, I/O equation:

- a. $h[n] = \delta[n - 5]$
- b. $h[n] = u[n - 5]$
- c. $h[n] = (0.8)^n u[n]$
- d. $h[n] = (0.8)^n (u[n] - u[n - 8])$
- e. $h[n] = (0.8)^n u[n] + (-0.8)^n u[n]$
- f. $h[n] = (0.8j)^n u[n] + (-0.8j)^n u[n]$

Prob 5:

Using the z-transform, determine the system function $H(z)$ and the causal impulse response $h[n]$.

- a. $y[n] = 0.8y[n - 1] + x[n]$
- b. $y[n] = 0.8y[n - 1] + x[n] + x[n - 1]$
- c. $y[n] = 0.8y[n - 1] + 0.25y[n - 2] + x[n]$
- d. $y[n] = 0.9y[n - 1] - 0.2y[n - 2] + x[n] + x[n - 1] - 6x[n - 2]$

Prob 6:

Calculate the convolution $x * y$

$$x[n] = 2^n u[n], \quad y[n] = \cos(\pi n) u[n]$$

Prob 7:

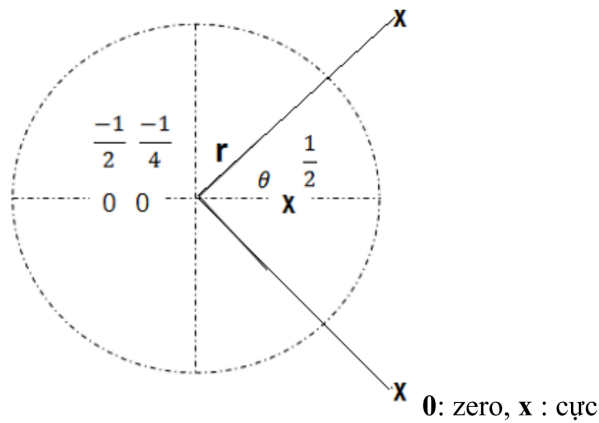
Consider the sequence $x[n]$ with z-transform

- a. $X(z) = \frac{3(1+0.3z^{-1})}{1-0.81z^{-2}}$
- b. $X(z) = \frac{6z^{-1}}{(1-z^{-1})(1-0.25z^{-1})}$

Using the inverse z transform, determine all possible $x[n]$ and the corresponding ROC and the corresponding stability-causality properties of the system.

Prob 8:

A system function $H(z)$ of an LTI system has pole-zero diagram as follows, where the gain $G = \frac{1}{2}$, $\theta = \frac{\pi}{4}$, $r = \sqrt{2}$:



- Determine the system function $H(z)$ of the system.
- Determine all possible impulse responses $h[n]$ of the system for all values of n .

Prob. 9:

Given the LTI system with the system function

$$H(z) = \frac{4 + z^{-1} + 3z^{-2}}{2 - 5z^{-1} + 3z^{-2}}$$

Find the stable impulse response.

Reference: For DSP learning materials, visit <https://www.mathworks.com/help/dsp/>.